

DIVERSITY AMONG ANIMALS

Chapter

9

Major Concept

In this Unit you will learn:

- Characteristics of Animals
- Criteria for Animal Classification
- Invertebrates
- Chordates





Diversity Among Animals

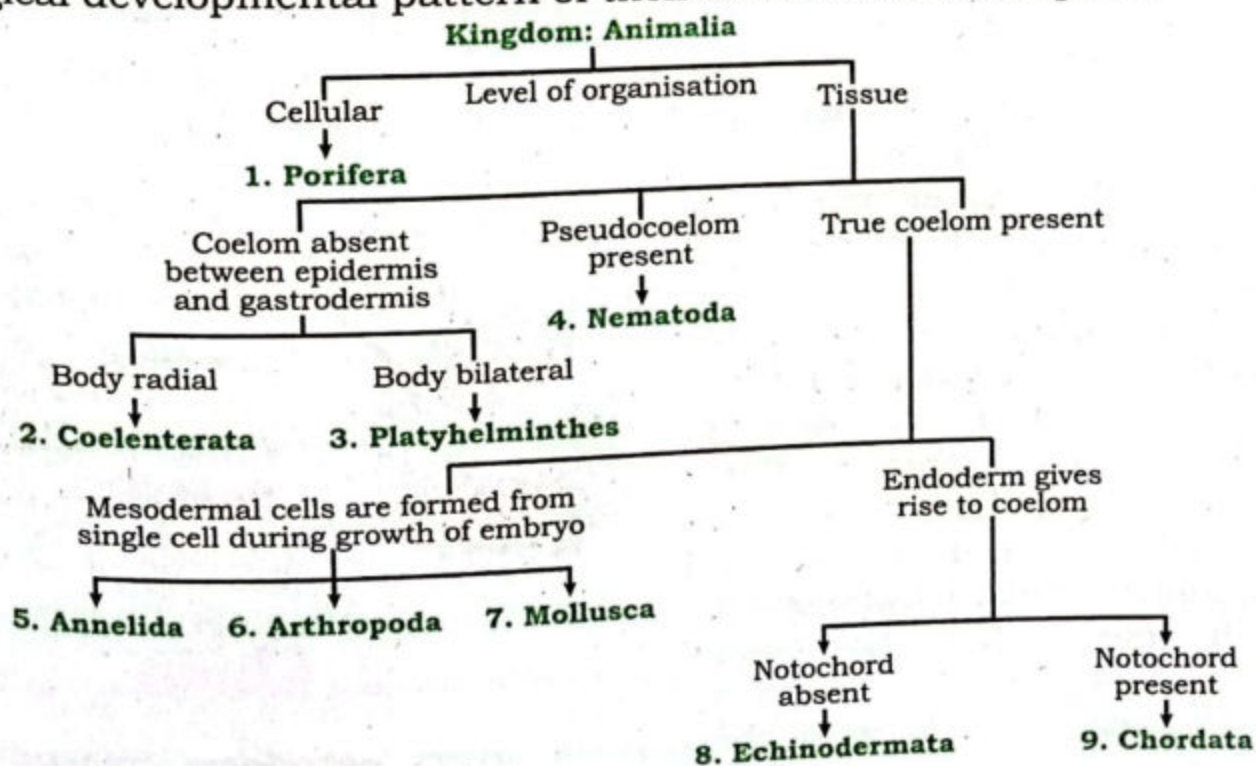
The term animal derived from Latin word "anima" means breath or soul. Animals are incredibly diverse in structure they range from simple parazoan that can only be seen with a microscope to blue whales, which weigh in upto 150 tons and 40m long. A good definition of an animal is not as easy as it might first appear. There are exceptions to nearly every criterion for distinguishing an animal from other life forms. However, when taken together, the following characteristics of animals will serve our purposes.

9.1 CHARACTERISTICS OF ANIMALS

Animals are multicellular, heterotrophic (ingestive) eukaryotes that can reproduce sexually. Although animal species exhibits a tremendous diversity of reproductive styles, all are capable of sexual reproduction. Lack cell wall, motile and able to make responses to external stimuli.

9.2 CRITERIA OF ANIMAL CLASSIFICATION

The kingdom animalia is divided into 33 groups called phyla. But we will consider in depth only nine major phyla. These major phyla are Porifera, Cnidaria, Platyhelminthes, Aschelminthes, Annelida, Mollusca, Arthropoda, Echinodermata and chordata whereas rest of the groups are minor phyla. This classification or grouping of animals is called taxonomy or systematics. It is carried out primarily on the basis of their evolutionary relationships. Clues to these relationships are found in (i) Comparative morphology and (ii) internal architecture which includes presence and absence of tissues, number of tissue layers, symmetry and the embryological developmental pattern of their coelom and blastopore.



9.2.1 Classification based on presence and absence of tissues

All animals are multicellular and their cells are eukaryotic, these cells are joined together into tissues, tissues into organs and organs system into organism. One phylum Porifera is grouped in a separate sub-kingdom **Parazoa**, because its members lack true tissues. Rest of the eight phyla consisting sub-kingdom eumetazoa (true metazoans) have tissues organized into organs (in lower groups) and organs into systems (in higher forms).

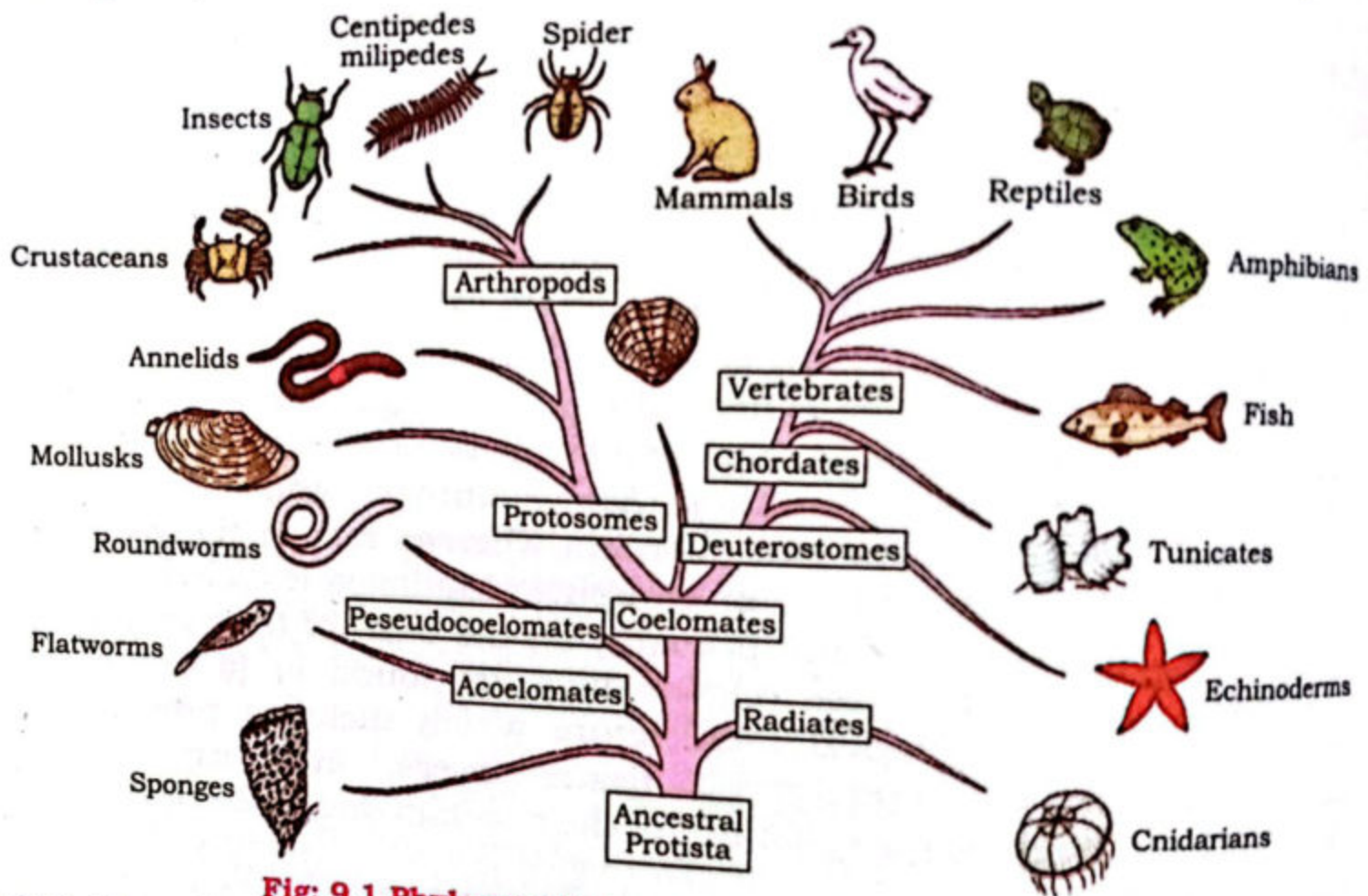


Fig: 9.1 Phylogenetic tree of the animal kingdom

9.2.2 Classification based on number of tissue layers

One of the main events during the development of animals is the establishment of germ layers or tissue layers. From which all other structures are derived. Although a total of three germ layers is seen in most animal embryos. Some animals only have two germ layers: **ectoderm** and **endoderm** called **diploblastic**, such animals have the tissue level of organization example; cnidarians. Animals with three

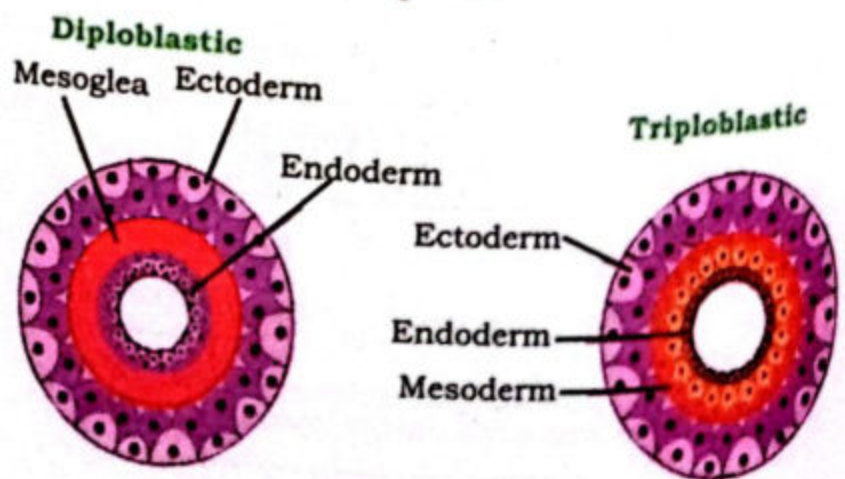


Fig: 9.2 Diploblastic and Triploblastic organization

germ layers **ectoderm**, **mesoderm** and

endoderm called **triploblastic** such animals have organ and system level of organization. Example: Platyhelminthes to chordata.

9.2.3 Classification based on symmetry

The simplest animals, the sponges have many representatives whose bodies are irregular, and variable in shape so called asymmetrical. The eumetazoans are divided into two major groups on the basis of body symmetry. Members of phylum cnidaria have **radial symmetry** and group called **Radiata**. A radial animal has a top and bottom, or an oral and aboral side, but no head end and rear end and no left and right. The other major groups of eumetazoa are **bilateral symmetry**. A bilateral animal has not only a dorsal (top) side and ventral (bottom) side, but also an anterior (head) end and a posterior (tail) end and a left and right side. Animals of this group collectively called the **bilateria**.

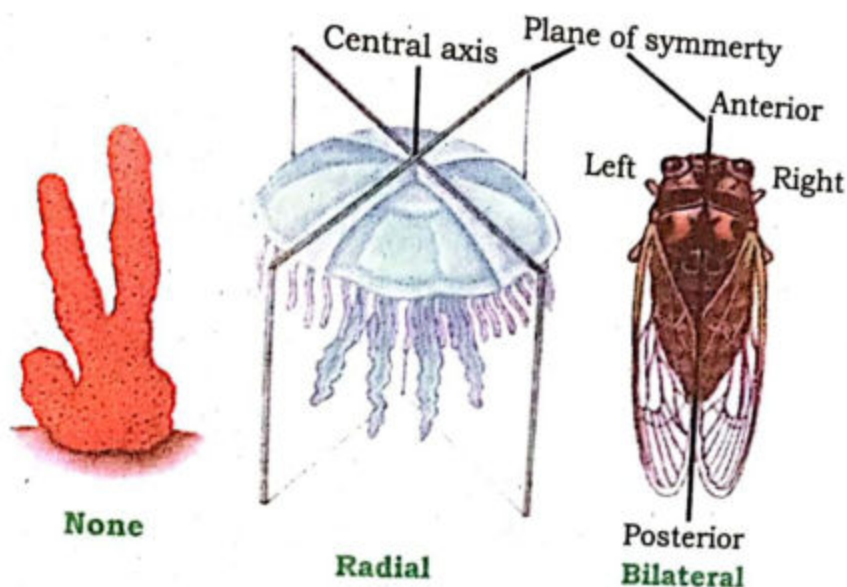


Fig: 9.3 Symmetry in animals

9.2.4 Classification based on coelom

The bilaterally symmetrical animals are classified into three groups on the basis of body cavity or coelom. **Acoelomate**; simple animals such as flatworms lack any internal space or coelom between their body wall and their gut.

Pseudocoelomates; animal such as round worms (aschelminthes) have a cavity through it is incompletely lined with mesoderm.

There is a layer of mesoderm beneath the body wall but not around gut. **Coelomates**; animals from annelids to chordates have a true body cavity or coelom that is completely lined with

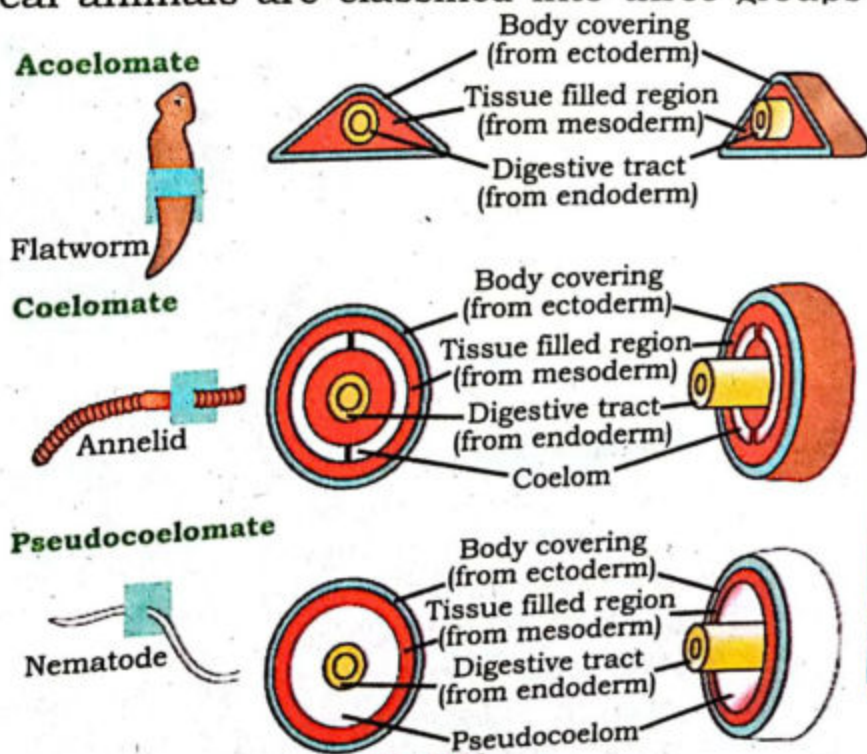


Fig: 9.4

Acoelomate, pseudocoelomate and coelomate

mesoderm. Coelomates are either protostomes or deuterostomes, when the blastopore (the site of invagination during development) is associated with mouth the animal is a **protostome** (Annelids, Molluscs and Arthropods). When the blastopore is associated with the anus and a second opening becomes the mouth, the animal is a **deuterostome** (Echinoderms, hemichordates and chordates).

A total of 1.3 million species of animals are placed in one of two major categories; **vertebrates**, those with a backbone, or vertebral column, and **invertebrates**, these lacking backbone. The invertebrates include 97% of all the animal species on earth.

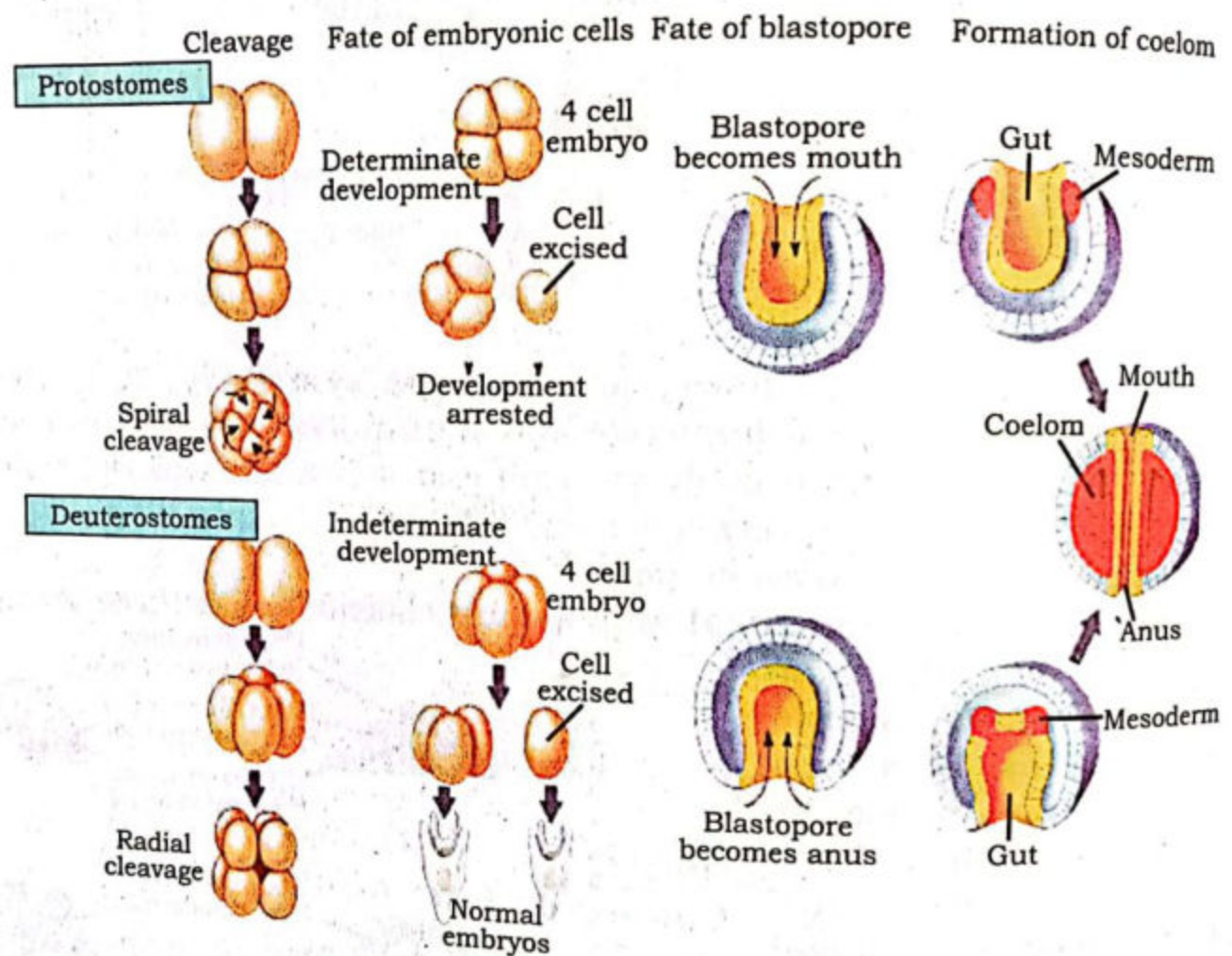


Fig: 9.5 Protostomes and Deuterostomes

9.3 PHYLUM PORIFERA (PORE BEARING)

General characters of phylum porifera

Sponges are the simplest animals lacking true tissues and organs. All sponges are aquatic filter feeders. The body of a simple sponge resembles a sac perforated with pores (holes). Water is drawn through the pores (**ostia**) into a central cavity the **spongocoel** then flows out of the sponge through a large opening called the **osculum**. The body of sponge consists of two layers of cells separated by a gelatinous mesenchyme. Outer is made up of flattened epithelial cells called **pinacocytes**, inner layer is



made up of flagellated cells called **choanocytes** or **collar cells**. Sponges may grow to over a meter in height and have an internal skeleton, which is composed of **spicules**, provides support for the body, the spicules may be formed from calcium carbonate, silica and protein. The bath sponge now rarely used which have proteinaceous endoskeleton.

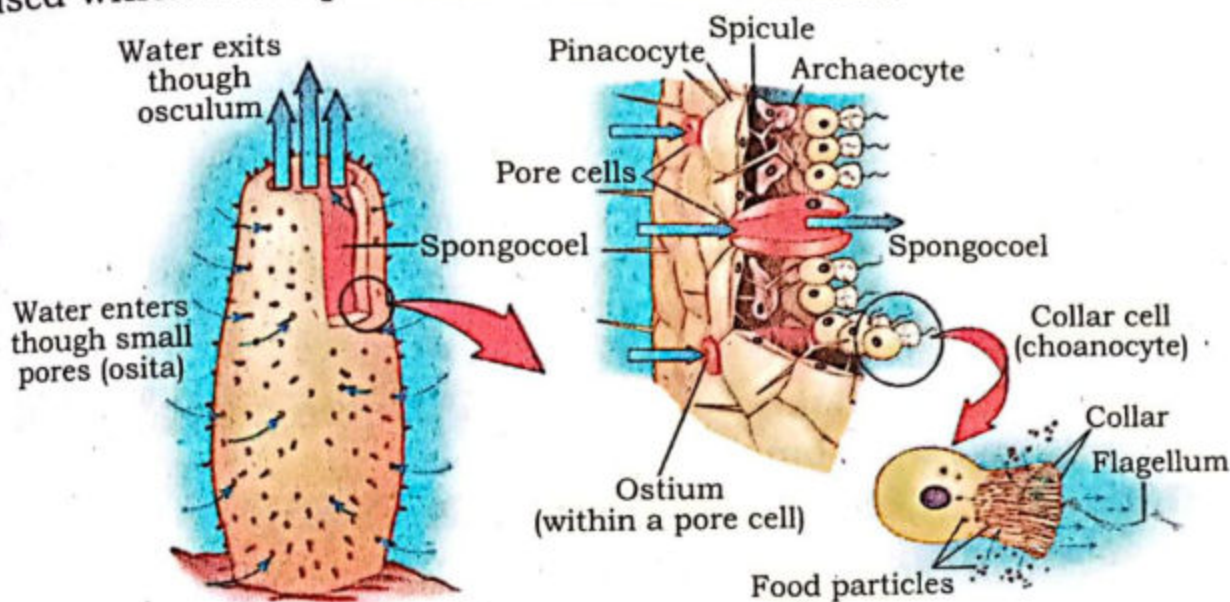


Fig: 9.6 Sponge cellular structure

Most sponges are hermaphrodite whereas in few members sexes are separate. Sponges can reproduce asexually by **fragmentation** or by **budding** or **gemmules**. During sexual reproduction eggs and sperm formed by **amoebocytes**. The sperm are carried out by water current to neighbouring sponges where fertilization takes place. The fertilized egg develops into a multicellular free-swimming larva which settles to the bottom and grows into an adult. Common sponges; *Sycon* (Fig: 9.7-a), *Euplectella* (Fig: 9.7-b) and *Spongilla* (Fig: 9.7-c)



Fig: 9.7: (a) Sycon

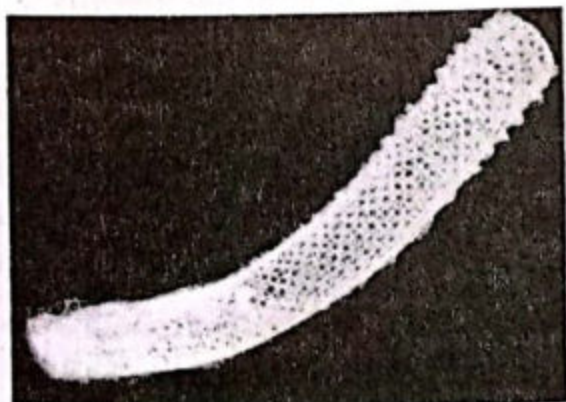


Fig: 9.7: (b) Euplectella



Fig: 9.7: (c) Spongilla

Extra reading material

Sponges are master filter. They can filter an amount of water 100,000 times their size each day. That means a basketball-sized sponge could filter an entire residential pool in one day. Sponges often use chemicals to deter predators from eating them. Scientists have discovered that some of these chemicals may have potential to treat cancer and HIV.

9.3.1 Evolutionary Adaptations in sponges

Sponges evolve from colonial flagellated protoctists that lived over 700 million years ago, these protoctists were probably related to choanoflagellates.

Digestion: The simplest form of digestion is found in sponges, which lack digestive tract. In sponges digestion is entirely intracellular.

Respiration: Respiratory system is not found in sponges. Diffusion is sufficient for gases exchange.

Excretion: Diffusion helps in excretion hence there is no special organ for the removal of nitrogenous waste.

Transport: Diffusion is sufficient for transport within sponges.

Nervous system: Sponges lack nerve cells.

9.3.2 Importance of sponges

Sponges with fibrous skeleton are used for the purpose of bathing, polishing, washing and scrubbing.

9.4 PHYLUM CNIDARIA (WITH NEMATOCYTES)

General characters of phylum cnidaria

The cnidarians are clearly a step above the sponges in complexity, their cells are organized into distinct tissues, including contractile tissues that act like muscle. Cnidarians all are aquatic majority of them are marine whereas few live in fresh water. They are radially symmetrical and diploblastic their body wall encloses a hollow cavity the **gastro-vascular cavity** or **coelenteron** hence the named coelenterata. All cnidarians are **carnivorous**, they paralyse or kill their prey with the help of special stinging cells called **cnidocytes** hence the name cnidaria. They have two distinct body forms polyp and medusa. **Polyps** are cylindrical with mouth and tentacles upward, **Medusa** on the other hand umbrella shaped whose mouth and tentacles downward. Cnidarian can reproduce by both asexual and sexual methods, asexual reproduction takes place by budding and regeneration whereas sexual reproduction involves the fusion of sperms and eggs released into the water, the fertilized egg developed into free swimming larva.

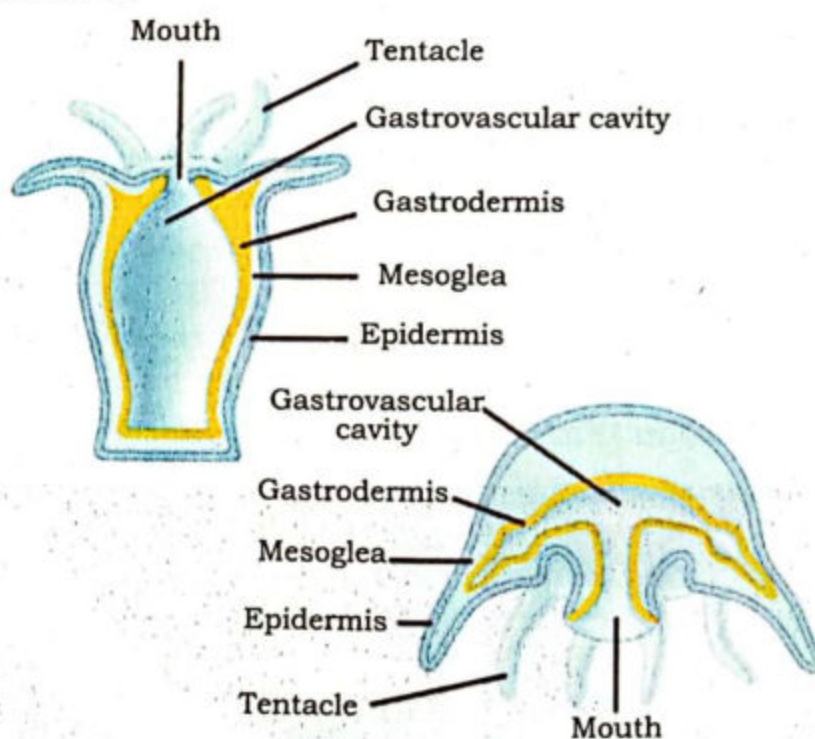


Fig. 9.8: Polyp and medusae



Common cnidarians: *Hydra* (Fig: 9.9-a), Jelly fish (Fig: 9.9-b), and Sea anemone (Fig: 9.9-c).



Fig: 9.9: (a) *Hydra*



Fig: 9.9: (b) Jelly fish



Fig: 9.9: (c) Sea Anemone

9.4.1. Evolutionary adaptations of cnidaria

Cnidarian evolved along one of the three evolutionary lines from the protoctista

Digestion: Digestive system consisting of a sac called gastro-vascular cavity with a single opening mouth help in both ingestion and egestion.

Respiration: There is no respiratory system both ectodermal and endodermal cell exchange gases with water.

Excretion: There is no excretory system, diffusion is only responsible for removal of nitrogenous waste.

Transport: Gastro vascular cavity help in transport of nutrients throughout the body.

Nervous system: The nerve cells are organized into tissue called a nerve-network or diffuse type nervous system.

9.4.2. Importance of Cnidaria

Coral reefs which are usually restricted to warm shallow water provide a heaven to large number of marine species. Jewelry and other decorative items are carved from red corals. **Red corals** named **MARJAN** are used by Hakeem in preparing eastern medicine.

9.5 PHYLUM PLATYHELMINTHES (FLAT WORMS)

General characters of phylum cnidaria

Flat worms are so named because their bodies are thin between the dorsal and ventral surfaces (flattened dorsoventrally). They range in size from nearly microscopic free living to 20 m long tapeworms. They are **acoelomate**, **bilaterally symmetrical**, **triploblastic** animals with organs

Extra reading material

Some cnidarians are the longest animals in the world, one of them that can stretch as long as 40m is the *Praya dubia*.

Jelly fish are made up of more than 95% water and they have no brains, blood or nervous system. Jelly fish can have up to 800 tentacles.

and organ system. Some flat worms have a head with a pair of **eye spots** that detect light, free living flat worm move with the help of cilia present on the ventral epidermis. Flatworms can reproduce by both asexual and sexual methods. Free living form reproduces asexually through regeneration. All forms can reproduce sexually and most are hermaphrodite (monoecious).

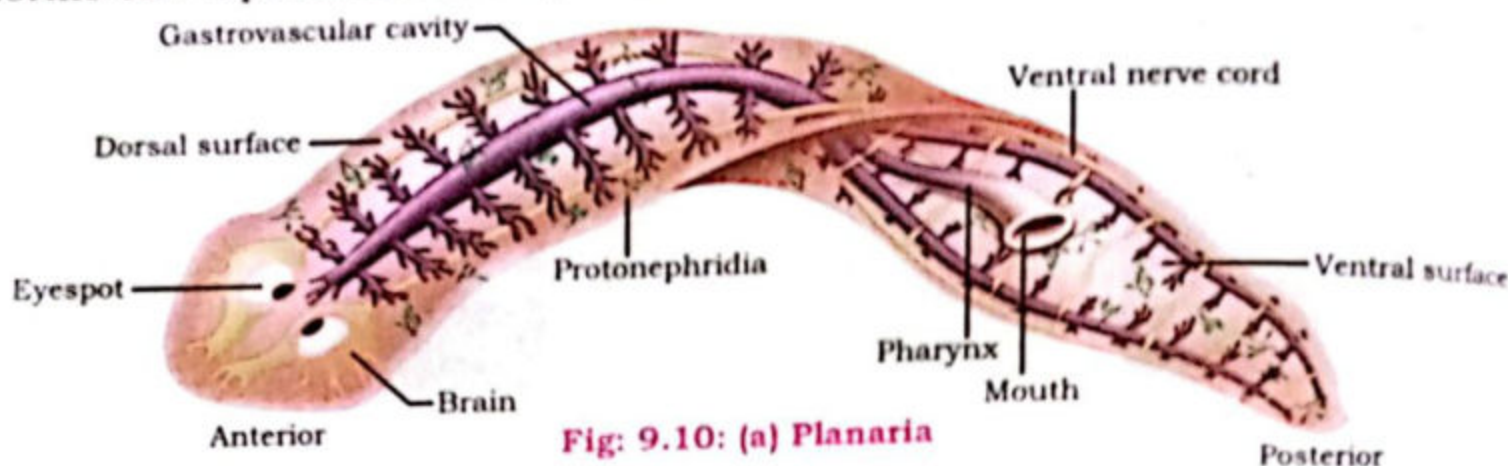


Fig: 9.10: (a) Planaria

Common Platyhelminthes: Planaria (*Dugesia*) (Fig: 9.10-a), Liver fluke (*Fasciola hepatica*) (Fig: 9.10-b) and tapeworm (*Taenia saginata*)



Fig: 9.10: (b) Liver fluke

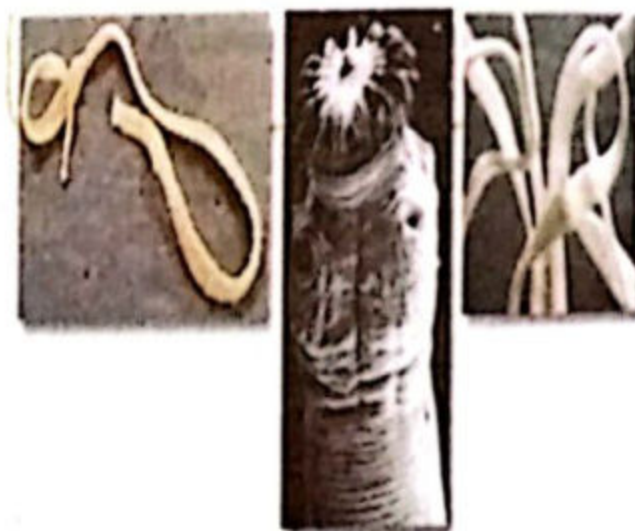


Fig: 9.10: (c) Tape worm

9.5.1. Evolutionary adaptations in Platyhelminthes

Flatworms do not look anything like cnidarians, the two share some features that have led biologist to speculate that they evolved from a common ancestor. Both have a gastro-vascular cavity.

Digestion: The food is digested in an intricately branched gastro-vascular cavity that distributes nutrients to all parts of the body.

Respiration: Exchange of gases by diffusion between the cells and the environment.

Excretion: Simple system of excretion consisting on network of canals ending in bulb shaped cells having cilia called flame cells (Proto-nephridia).

Transport: No specialized circulatory system.



Nervous system: Ladder type nervous system consists of the two lateral nerve cords, plus the connecting nerves looks like a ladder. Paired ganglia (collection of cells) function as brain.

Extra reading material

The flatworms are not really flat and have a rounded body instead. They do have a flat belly through. The flatworms can either be extremely tiny or can grow to become several inches long. They are pointed on both ends are characterized by eye spots on the head. Their colors may vary from being translucent white, green as well as brown. It could also be blotchy and mottled as well.

9.5.2 Importance of Platyhelminthes

Flatworms, such as the planarians, are mostly free living while some are parasites.

9.6 PHYLUM ASCHELMINTHES (NEMATODS)

General characters of phylum Aschelminthes

Aschelminths commonly called **round worms** are the most abundant animals on earth, found everywhere in sea, freshwater and soil. Round worms have a varied mode of existence from free living, scavengers to predators and parasitic on animals and plants. Most are microscopic but some parasitic form reach a meter in length, have long bilaterally symmetrical, triploblastic, pseudocoelomate and cylindrical bodies with pointed ends. The muscles of nematodes are all longitudinal and their contraction produces a thrashing motion. Round worm reproduce sexually the sexes are separate in most species females generally larger than males, fertilization is internal. Common

Aschelminths: *Ascaris* (Fig: 9.11), *Ancylostoma* and *Wuchereria*.



Fig: 9.11: Male and Female Ascaris

9.6.1 Evolutionary adaptations in Aschelminthes

Digestion: Tubular gut running from mouth to anus. Digestive system is complete.

Respiration: Diffusion sufficient for gases exchange, has no respiratory system.

Transport: Circulatory system not found.

Excretion: Excretory system consists of two canals, which run down the length of worm.

Nervous system: Sensory organs in the head transmit information to a simple brain composed of a ring of ganglia.

Extra reading material

Nematodes are essential in the nitrogen cycle and for regulating the decomposition of organic matter. One species of nematode is known to control mosquitos by eating their larvae.

There are 60 billion nematodes for every human on earth. According to Barnes 80% of the living animals on earth are nematode worms

9.6.2. Importance of Aschelminthes

Round worms are important as agriculture pest that attack the roots of plants. Human serve as host of 50 species.

Extra reading material

Pest is any organism that spread any disease, causes destruction or is otherwise a nuisance. For example, mosquito, rodents and weeds.

9.7. PHYLUM ANNELIDA (SEGMENTED WORMS)

General characters of Phylum Annelida:

Annelids are commonly called segmented worms, have the most complex body. Structures among of all the worms, they live on land, in moist soil, in freshwater or in sea. Many annelids are active free-swimming predators, some are aquatic filter feeder living in tubes buried in mud whereas leeches are ectoparasites, they are all triploblastic, bilaterally symmetrical, coelomate and protostomes with an organ system level of body organization. Chitinous chaetae also called **setae** with or without **parapodia**

are usually present in the most of annelids and help in locomotion. Reproduction is usually sexually, most of annelids are hermaphrodite but in few sexes are separate, development through trochophore larvae. Common Annelids: Earthworm (Fig: 9.12-a), Neries and Leech (Fig: 9.12-b).

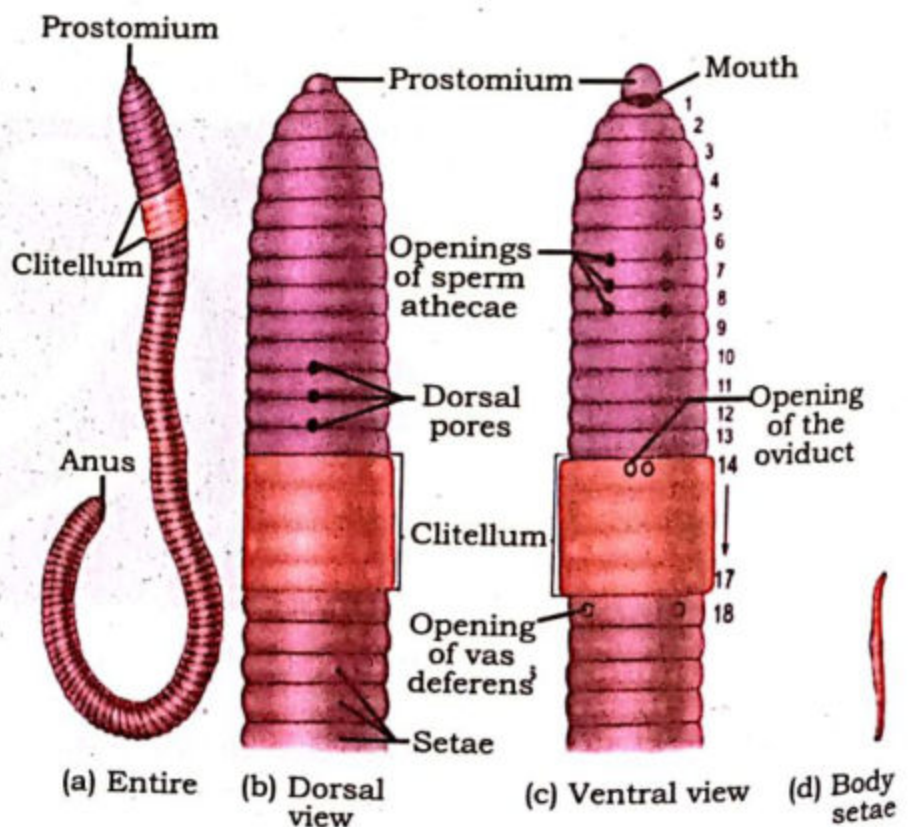


Fig: 9.12 (a) Earth worm



9.7.1. Evolutionary adaptations in Annelids

Annelids are supposed to have evolved from a primitive flat worm like ancestor in the sea.

Digestion: Tubular digestive system with both mouth and anus, complete system.

Respiration: Respiratory system is not found, diffusion is sufficient for gases exchange.

Transport: Closed type circulatory system with blood confined to the heart and blood vessels.

Excretion: Excretory organs are nephridia which are found in most segments.

Nervous system: Nervous system consists of simple ganglionic brain in the head.

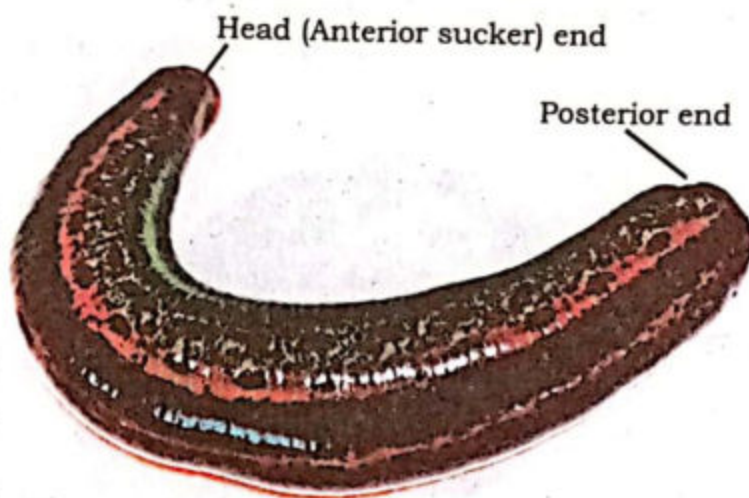


Fig: 9.12 (b) Leech

Extra reading material

Leeches have been used for medicinal purposes for a very long time. Back then, it was believed that leeching would get rid of the "Bad Blood" that caused diseases.

9.7.2 Economic Importance of Annelids

Earth worm help the farmer by continuously ploughing the soil and adding nitrogenous waste into it, thus making soil fertile, they are also used as fish bait. Leeches are ectoparasite and suck the blood.

9.8. PHYLUM MOLLUSCA (SOFT BODIED)

General characters of phylum Mollusca

Molluscs are soft bodied animals, but most are protected by hard shell made of calcium carbonate, slug, squids and octopus have reduced shell, most of which are internal, or they have lost their shell completely during evolution. All the molluscs are triploblastic, bilaterally symmetrical, coelomates and protostomes with organ system grade of body organization. All molluscs have a similar body plan with

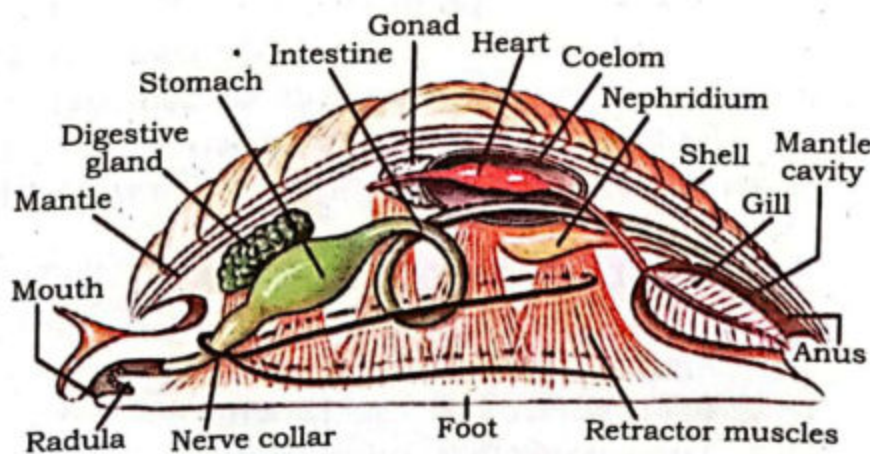


Fig: 9.13 Anatomy of snail

three main parts, **Muscular foot** usually used for movement, **visceral mass** containing most of the internal organs and a **mantle** a fold of tissue that drapes over the visceral mass and secretes a **shell**. Many molluscs feed by using a strap like rasping organ called a **radula** to scrape a food. Reproduction is always sexual, some species have separate sexes and others are hermaphrodite, they all pass through a trochophore larvae stage. Common Molluscs: *Unio* (Fig: 9.14-a), *Octopus* (Fig: 9.14-b) and *Pearl oyster*

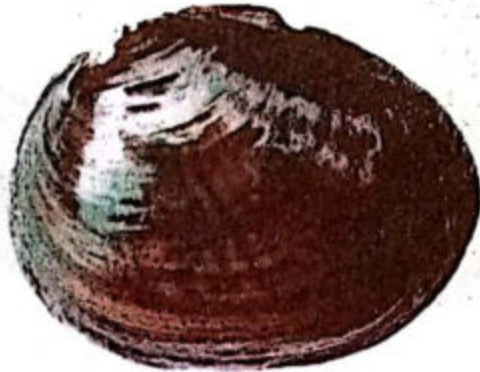


Fig: 9.14 (a) *Unio*



Fig: 9.14 (b) *Octopus*

Extra reading material

The oyster is usually ambisexual. It begins life as a male and then becomes a female. It may go back and forth many times. The giant squid is the largest creature without backbone. It weighs up to 2.5 tons and grows up to 60 feet long.

9.8.1. Evolutionary adaptations in molluscs

Digestion: Digestive system is straight with mouth and anus.

Respiration: Exchange of gases in aquatic forms with gills and simple lungs in terrestrial form.

Transport: Open type circulatory system in which blood directly bathing the organs in a haemocoel.

Excretion: well developed excretory system using tubular nephridia.

Nervous system: Nervous system consists of ganglia connected by nerves, but many more of the ganglia are concentrated in the brain.

9.8.2. Economic importance of mollusca:

A variety of molluscs called shell fish, together with crustaceans are still an important source of food. Their shells are decorative, some members make valuable pearl.

9.9. PHYLUM ARTHROPODA

General characters of phylum Arthropoda

Arthropoda is the largest phylum of the animal kingdom, they are found everywhere on earth wherever the life is possible, even in the oil wells. Arthropods are bilaterally symmetrical, triploblastic, coelomates and protostomes. The diversity and success of arthropods are largely related to their segmentation, hard exoskeleton, and jointed appendages. The body of



arthropods is completely covered with cuticle, an exoskeleton, made of protein and chitin. The exoskeleton protects the animals and provides point of attachment for the muscles that move the appendages. Sexes are usually separate and metamorphosis is common occurrence.

Common Arthropods: *Scorpion* (Fig: 9.15-a), *Prawn* (Fig: 9.15-b), *Mosquito* (Fig: 9.15-c), *Honey bee* and *House fly* etc.



Fig: 9.15 (a) Scorpion



Fig: 9.15 (c) Mosquito



Fig: 9.15 (b) Prawn

Extra reading material

Arthropods exist in the harshest conditions in the world, from very cold places to some of the hottest in the world. A scorpion, which is an arthropod, can survive even after being frozen solid.

9.9.1 Evolutionary adaptations in arthropods

Digestion: Alimentary canal is well developed with mouth and anus. Mouth is assisted by jaws, digestive system is complete.

Respiration: Arthropods efficient gas exchange is accomplished by gills in aquatic forms such as crustaceans, and by either tracheae or book lungs in terrestrial forms.

Transport: Arthropods have well developed circulatory system, blood not only travels through vessels but also empties into the **haemocoel**, where it bathes the internal organs directly this arrangement known as an **open type circulatory system**.

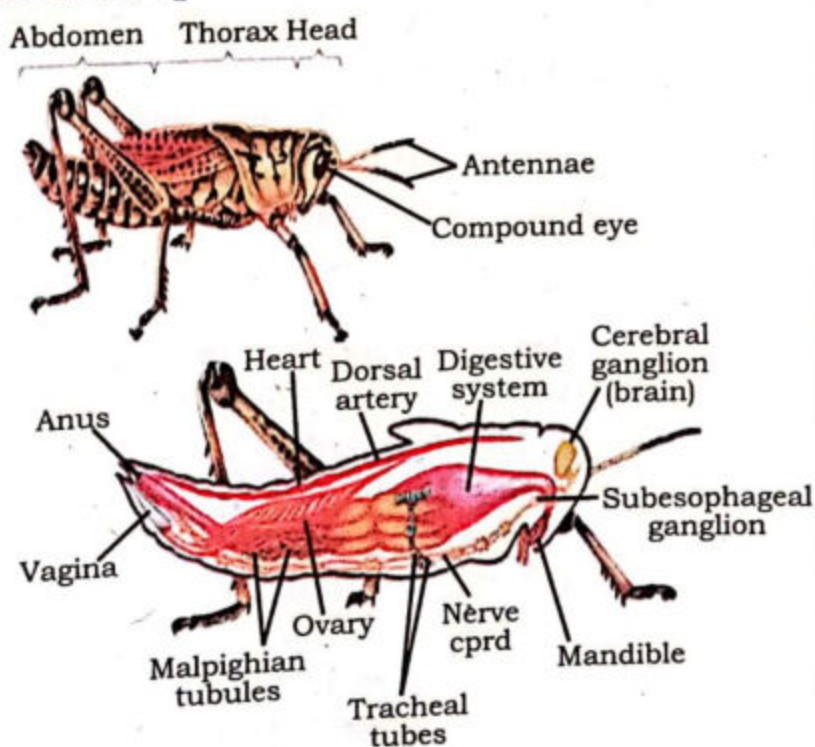


Fig: 9.16 Internal structure of arthropods

Excretion: Excretory organs are mostly **malpighian tubules**. They remove nitrogenous waste from the **haemolymph** and also function in osmoregulation.

Nervous system: Arthropods nervous system is similar in plan to that of annelids but more complex. It consist of a brain composed of fused ganglia in head, connected to a series of ganglia running down the length of the body and linked by a ventral nerve cord.

9.9.2 Economic importance:

Arthropods are of great economic importance, the predominant group of arthropods, the insects not only helps in pollination but also predator on plant pest. Many cause diseases in plant and animals, by transmitting bacteria and viruses. In human beings they are responsible for the transmission of trypanosoma, plasmodium and germs of cholera. Arthropods are an important source of food for many animals and carnivores plants. Sea food that is not fish or molluscs is generally arthropods.

9.10. PHYLUM ECHINODERMATA (SPINY SKINNED)

General characters of phylum Echinodermata (Gr; Echinus = spiny and dermos = skin)

The representative of phylum Echinodermata are exclusively marine animals, they are radially symmetrical as adult, their larva is a free swimming, filter feeder with bilateral symmetry. Adult echinoderms lack head, brain and segmentation. They are triploblastic, coelomates and deuterostomes with organ-system grade of body organization. The body is covered over by a delicate epidermis stretched over a firm endoskeleton of fixed or moveable calcareous plates with spines. Echinoderms move on numerous tiny tube feet, delicate cylindrical projection that extends from lower surface of the body. Tube feet are part of water vascular system, which functions in locomotion, respiration and food capture.

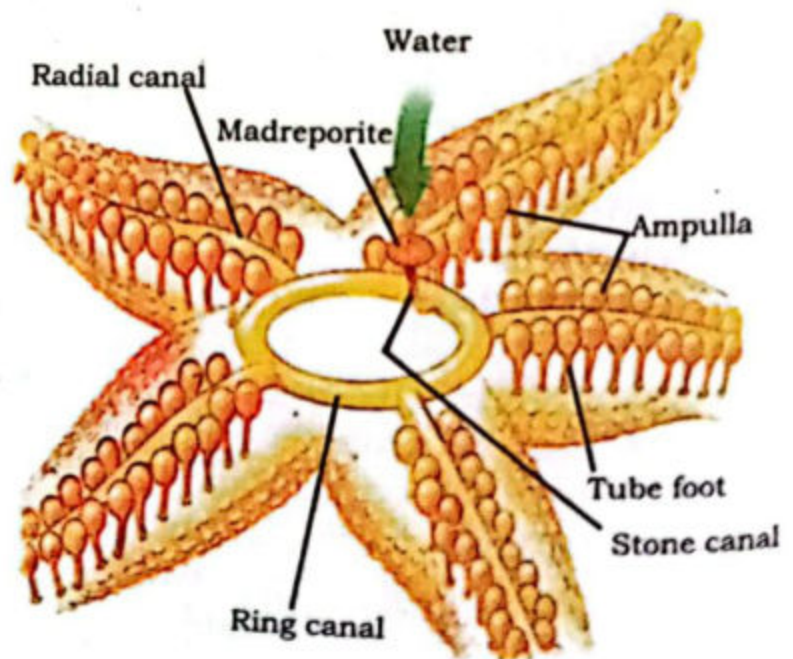


Fig: 9.17 Water vascular system

Extra reading material

Some echinoderms are vital for the ecology of the world's oceans. Sea cucumbers constantly sift through the bottom sand and absorb dead organic matter from it.



Most echinoderms reproduce by releasing sperms and eggs into the water, where fertilization occurs and a free-swimming larva develop, the sexes are usually separate.

Common echinoderms are Star fish (Fig: 9.18-a), Sea urchin (Fig: 9.18-b), Sea cucumber and Brittle star.

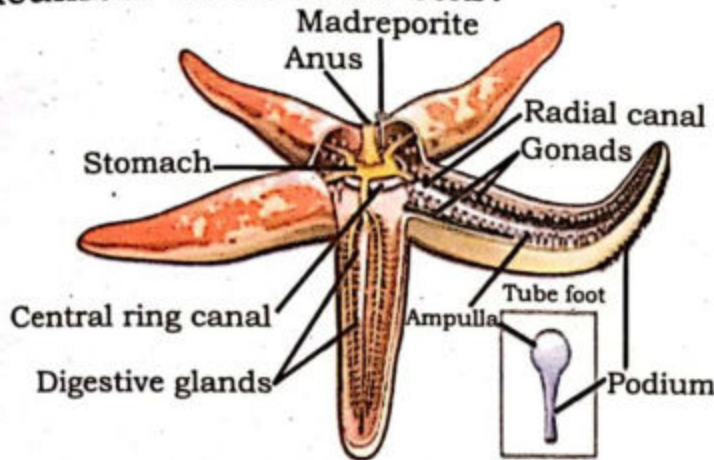


Fig: 9.18 (a) Star Fish

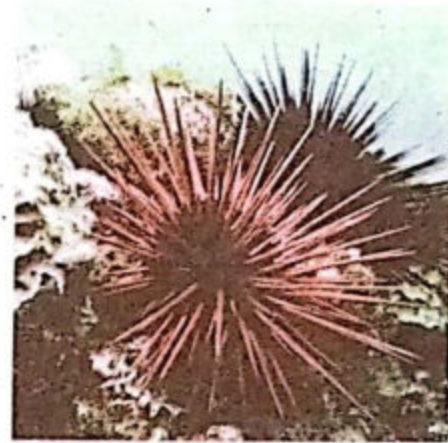


Fig: 9.18 (b) Sea urchin

9.10.1. Evolutionary adaptations in echinoderms

Digestion: Echinoderms have complete digestive tract but in some species as a brittle star it is incomplete.

Respiration: Gas exchange occurs through the tube feet and in some forms numerous tiny skin gills project through the epidermis, respiratory tree in sea cucumber.

Transport: The echinoderms lack a circulatory system, although movement of the fluid in their wall developed coelom.

Excretion: Echinoderm lack excretory organs, diffusion is responsible for loss of ammonia.

Nervous system: Echinoderms have a relatively simple nervous system with no distinct brain.

9.10.2. Economic importance of echinoderms

Echinoderms are efficient scavengers on seafloor, sea urchins that burrow into rocks and along a shore can accelerate the erosion of shorelines.

Extra reading material

PHYLUM HEMICHORDATA (Half notochord)

Hemichordates are exclusively marine, solitary or colonial, mostly tubicolous. Their body is soft, fragile, vermiform, unsegmented, bilaterally symmetrical, triploblastic coelomate and deuterostomes. Reproduction is mainly sexual. Sexes usually separate. Fertilization is external, development direct or indirect with free swimming tornaria larva.

Digestion: Digestive system complete with mouth and anus.

Respiration: Gas exchange through dorso-lateral pharyngeal gill slits.

Transport: Simple open and well developed circulatory system.

Excretion: Single glomerulus in the proboscis help in removal of nitrogenous waste.

Nervous system: Nervous system consisting mainly of subepidermal nerve plexus and dorsal collar nerve cord. **Common hemichordate:** Balanoglossus & Saccoglossus

9.11. PHYLUM CHORDATA

General characters of phylum Chordata:

Chordates in general, are bilaterally symmetrical, triploblastic coelomates and deuterostomes, animals having a complete digestive tract with a mouth and an anus. Coelom is well developed and internal organs are suspended in the coelomic cavity fixed by a thin membranous tissue called mesentery. They all as a rule, reproduce sexually. The chordates are extremely diverse group of animals united by four features that all possess at some stage of their lives.

Notochord: Flexible rod that extends the length of the body and provides an attachment site for muscles.

Nerve cord: Lying dorsal to the digestive tract, this hollow neural structure develops a bulging at its anterior end that becomes a brain.

Gill slits: located in the pharynx, these may form functional respiratory openings or may appear only as grooves during an early stage of development.

Post anal tail: most chordates have a tail extending posterior to the anus.

9.11.1 Classification of chordates

According to the recent classification chordates are classified into two groups.

- (i) Invertebrate chordates
- (ii) Vertebrates.

Group Invertebrate chordates (Protochordates)

The invertebrate chordates lack a head and backbone that is defining feature of vertebrates they are classified into two sub-phyla

- (i) Sub-phylum urochordata and
- (ii) Sub-phylum cephalochordates.

Sub phylum urochordata

Urochordates are also called tunicate because their body is enclosed in a sac called tunic. They are marine and mostly sessile. The tunic is provided with two openings an incurrent or buccal siphon and an excurrent or atrial siphon. Larva is motile and show chordate characters. Common urochordates: Herdmania (Fig: 9.20) and Ascidia

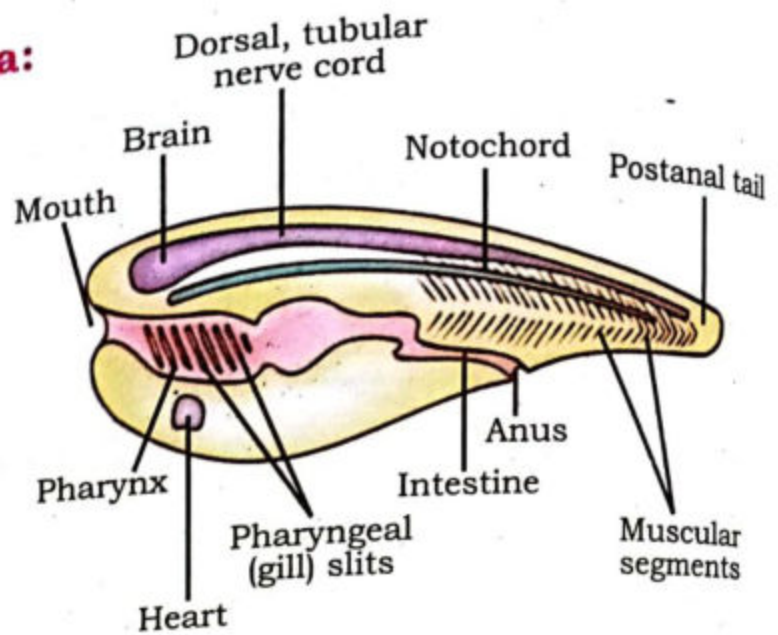


Fig: 9.19 Chordate body

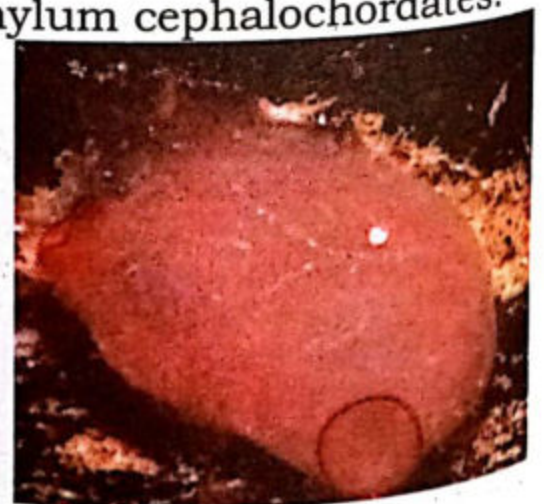


Fig: 9.20 Herdmania



Sub phylum Cephalochordata

The small fish like cephalochordates spend most of its time half buried in the sandy sea bottom filtering tiny food particles from the water. Common cephalochordate: Amphioxus (Fig:9.21)

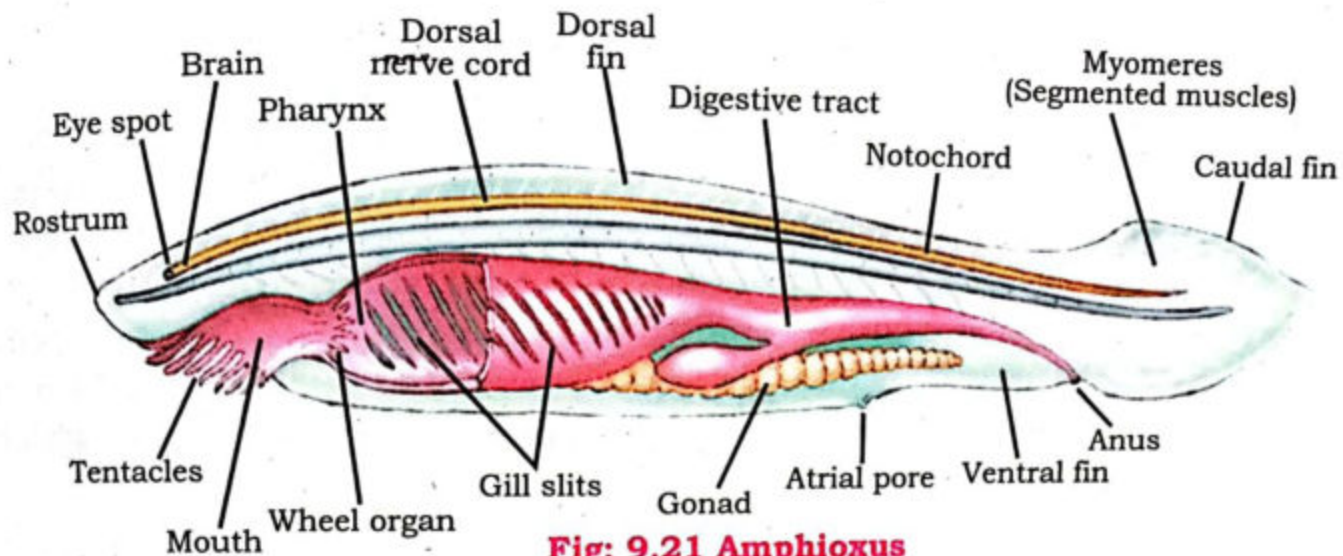


Fig: 9.21 Amphioxus

(ii) Group vertebrata

In the vertebrates notochord is usually replaced during development by a backbone or vertebral column. Vertebrates show other adaptations that have contributed to their successful invasion of most habitats. One is the presence of paired appendages. These first appeared as fins in fish and serve as stabilizers for swimming. Over millions of years, some fins were modified by natural selection into leg that allowed animals to crawl onto dry land, and later into wings that allowed some to take to the air. Another adaptation is an increase in size and complexity of their brain and sensory structures.

Classification of vertebrates

Vertebrates are classified into seven major classes

1. Class Jawless fishes (agnatha)

Small group of vertebrates which includes two groups of jawless fishes; the hagfishes and the lampreys, both have skeletons of cartilage and are eel like in shape. Both have unpaired fins located along the midline of the body. Both lack scales, and their smooth, slimy skin is perforated by circular gill openings.



Fig: 9.22 Lamprey

Evolutionary adaptations in agnatha

Hag fishes using pincer like teeth that surround the tongue to burrow into the coelomic cavity and ingest the prey's soft internal organs. Lamprey has sucker like mouth lined with teeth. Gill pouches or opening help in respiration.

2. Class Cartilaginous fishes. (Chondrichthyes)

The vertebrates of class Chondrichthyes, sharks, skates and rays are called cartilaginous fishes because they have relatively flexible endoskeleton made of cartilage rather than bone. Exoskeleton is made of enormous number of tiny sharp enamel coated denticles called **placoid scales**. Mouth is ventral in position and their tail fin is heterocercal. There are present many usually 5 exposed gill slits on side which are not covered over by a gill cover the operculum. Scoliodon (dog fish) is small shark which is common in our seas.

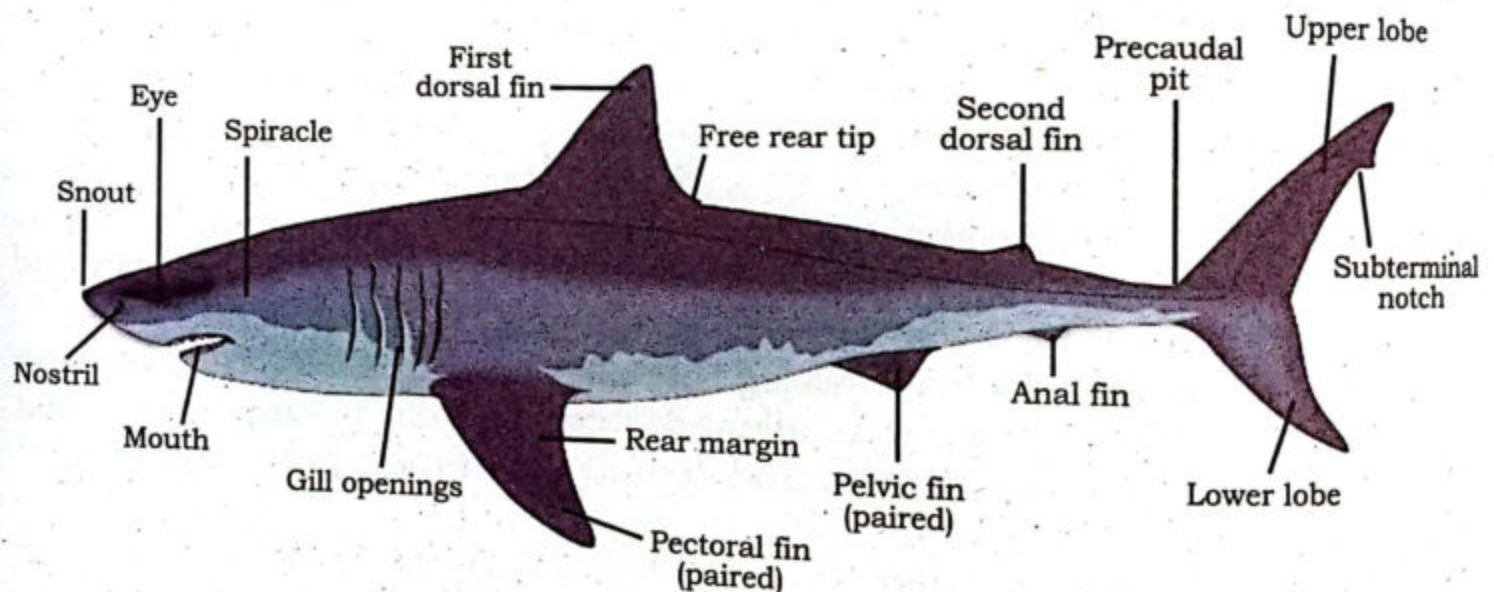


Fig: 9.23 Shark

3. Class Bony fishes (Osteichthyes)

Class osteichthyes (bony fishes) are the most numerous, both in individuals and in species.

Endoskeleton in these fishes is bony and the exoskeleton which is made up of tiny bony plates which are called **cycloid** or **ctenoid** scales according to whether their outer edge is smooth or spiny.

Mouth opening is present at the anterior tip. The gills are covered over on each side, by a gill cover called operculum. Most of these fishes have an air bladder which acts as a hydrostatic organ. Tail fin is usually **homocercal**

Extra reading material

The fastest fish is the sailfish. It can swim as fast as a car travels on the highway. The slowest fish is a seahorse. It swims so slowly that a person can barely tell it is moving.



or **diphycercal**. This group includes Eel, sea horse (Fig: 9.24-a), flying fish (Fig: 9.24-b), globe fish (Fig: 9.24-c), pomfret (Fig: 9.24-d) and Carps (rahu) (Fig: 9.24-e) etc.

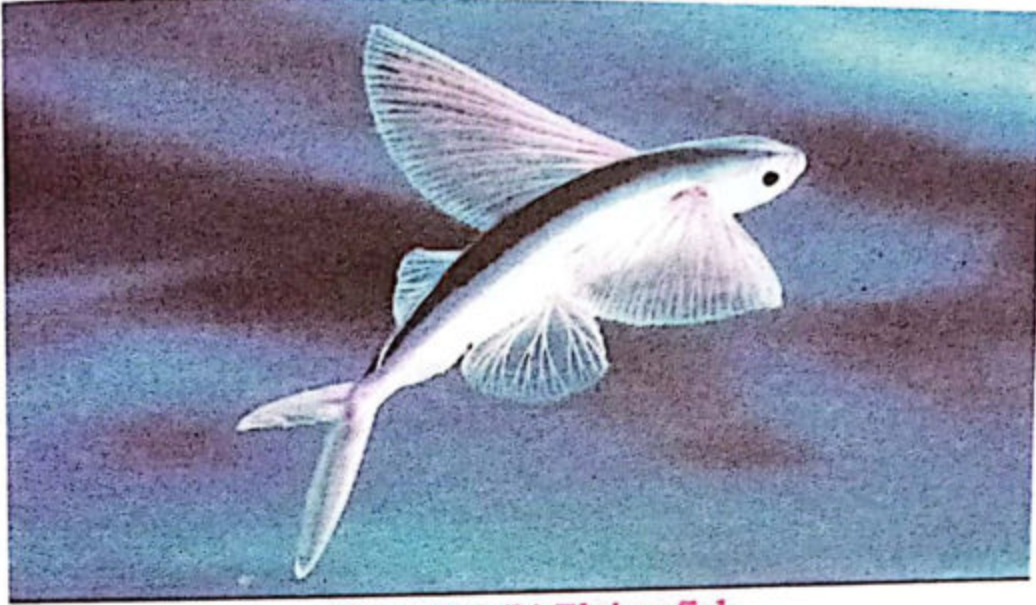


Fig: 9.24 (b) Flying fish



Fig: 9.24 (a) Sea Horse

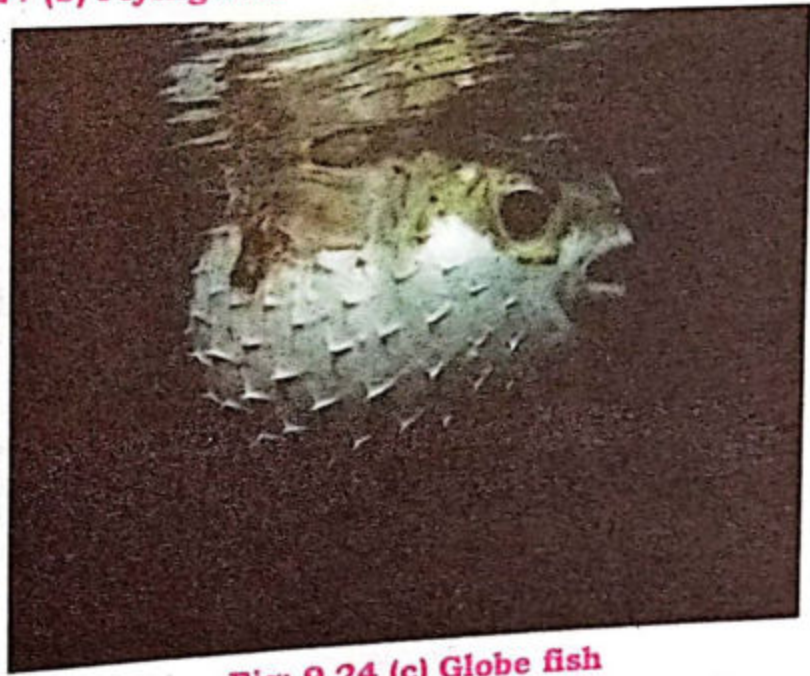


Fig: 9.24 (c) Globe fish



Fig: 9.24 (d) Pomfret



Fig: 9.24 (e) Carps

4. **Class Amphibia (Living at both places; water and land)**

The first vertebrates on land were members of class amphibia. The limbs of amphibians show varying degree of adaptation to movement in land, from the belly-dragging, crawl of salamanders (Fig: 9.25-b) to the efficient dive of frogs (Fig: 9.25-a). Lungs replace gills in most adult forms, and a three chambered heart circulates blood more efficiently. Skin of most of amphibians remains moist, since it serves as an additional respiratory organ that supplements poorly developed lungs. Amphibians are ectothermic depend on environment to regulate their body temperature. To avoid extremes of temperatures they undergo **hibernation** in winter and **aestivation** in summers.

Extra reading material

Unlike reptiles and mammals, amphibians don't have the ability to chew their food. They are also poorly equipped dentally, with only a few primitive "vomerine teeth" in the front upper part of the jaws that allow them to hold onto wriggling prey.



Fig: 9.25 (a) Frog



Fig: 9.25 (b) Salamander

5. **Class Reptilia (To crawl and creep)**

Reptiles evolved from an amphibian ancestor about 250 million years ago. Early reptiles the dinosaurs ruled the land for nearly 150 million years. Reptiles contain tough, scaly skin that resists water loss and protects the body. Fertilization is internal and egg is covered with shell. The shell prevents the egg from drying, while an internal membrane, the amnion, encloses the embryo in the watery environment that all developing animal embryo require. Reptiles respire through lungs, three chambered heart became modified, allowing better separation of oxygenated and deoxygenated blood. The limbs and skeleton evolved adaptations that provided better support and more efficient movement on land.

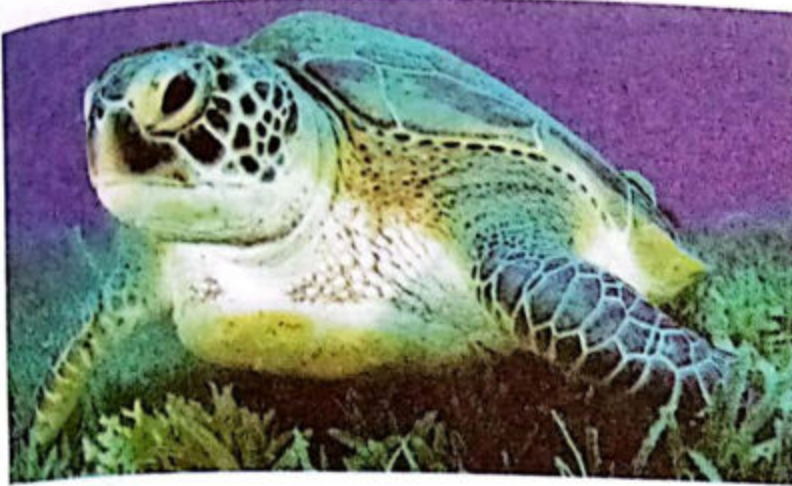


Fig: 9.26 (a) Green Turtle



Fig: 9.26 (b) Tortoise

Common reptiles found around the world are tortoise (Fig: 9.26-b) and turtles, lizards (Fig: 9.26-e), snakes (Fig: 9.26-d), crocodiles and alligators (Fig: 9.26-c). Many species of tortoises and turtles including the endangered green turtle (Fig: 9.26-a) the *chelone mydas* are found in Pakistan.



Fig: 9.26 (c)
Crocodile and alligator



Fig: 9.26 (d)
Snake



Fig: 9.26 (e)
Lizard

6. Class Aves (Birds)

Birds evolved from reptiles during Mesozoic era. Amniotic eggs and scales on the legs are just two of the reptilian features we see in birds. A bird can be defined as feather covered bipedal flying vertebrate possessing wings. Feathers which cover the body all over constitute a unique and basic identifying character of birds. Birds regulate their body temperature by physiological and behavioral mechanisms, maintaining a temperature that is usually higher than that of their surroundings. These animals have a high metabolic rate that increases their demand for energy and require oxygen for tissues. Four chambered heart keeps oxygenated blood separated from deoxygenated blood. Respiratory system of birds supplemented by air sacs that supply oxygenated air to the lungs. Forelimbs are modified into wings. They have hollow, very light bones laced with air cavities, horny beak has replaced jaws. Fertilization is internal and

eggs are large, amniotic and covered over by hard calcareous shells. The modern birds are divided into two group i.e. Ratitae (running birds) and carinatae (flying birds).

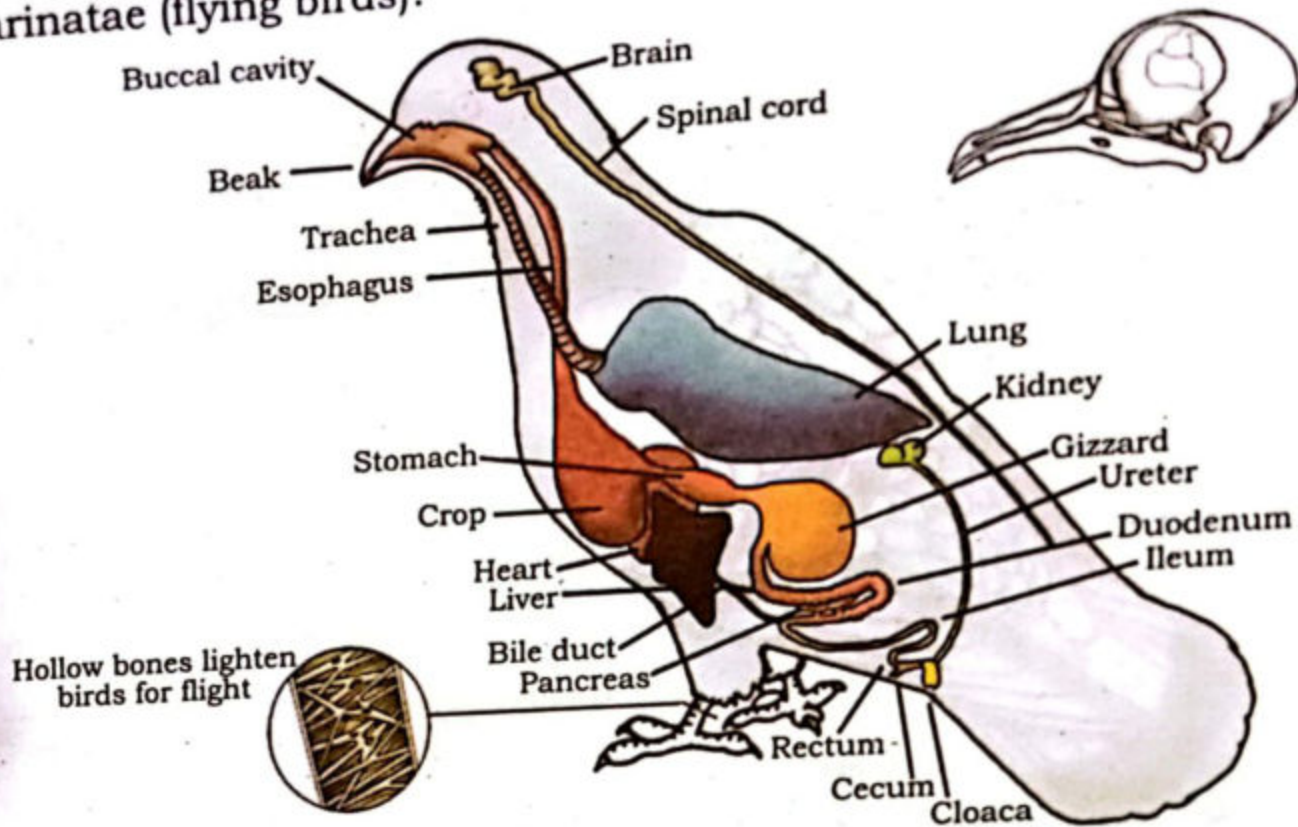


Fig: 9.27 Internal organs of birds

Sub class ratitae (sternum raft like)

These birds cannot fly because they have comparatively heavy weight and their wings are either vestigial or rudimentary. They have a flat sternum without keel and accordingly their flight muscles are poorly developed. The distribution of ratitae is also restricted. None of them is found in Pakistan. Common flightless birds are Ostrich (Fig: 9.28-a), emu (Fig: 9.28-b), cassowary, kiwi (Fig: 9.28-c) and penguin.



Fig: 9.28 (a) Ostrich



Fig: 9.28 (b) Emu



Fig: 9.28 (c) Kiwi



Sub class Carinatae (sternum with keel)

They are usually small, light weight birds whose wings are highly developed and feathers of their wings have an interlocking mechanism. Their sternum is provided with a crest like keel to accommodate the highly developed pectoral flight muscles. The flying birds are distributed all around the world. Common flying birds are pigeons (Fig: 9.29-a), crow (Fig: 9.29-b), parrot (Fig: 9.29-c), owl, peacock and chakor.



Fig: 9.29 (a) Pigeon



Fig: 9.29 (b) Crow



Fig: 9.29 (c) Parrot

7. Class Mammalia

The mammals first appeared approximately 250 million years ago and came into prominence after the extinction of the dinosaurs roughly 70 million years ago. Mammals are named for the mammary glands used by all female members of this class to suckle their young. Mammals have an active metabolism and are endotherms. Efficient respiratory and circulatory systems having four chambered heart support a high metabolic rate. A sheet of muscle called the diaphragm helps ventilate the lungs. Hairs and a layer of fat under the skin help the body retain metabolic heat. With the exception of the egg-laying monotremes, such as the platypus and spiny anteater, mammalian embryo develop in the uterus. Class mammalia is divided into three sub classes Prototheria, Metatheria and Eutheria, on the basis of the mode and developmental conditions of their new born babies.

Sub class Prototerhia (egg laying mammals)

The platypus (Fig: 9.30-a) and the spiny anteaters (Fig: 9.30-b) are the only living mammals that lay eggs. The egg which is reptilian in structure and development, contain enough yolk to nourish the developing

embryo. Monotremes have hair and produce milk, two of the most important features of Mammalia.



Fig: 9.30 (a) Platypus



Fig: 9.30 (b) Spiny Anteaters

Sub class Metatheria (pouched mammals)

Kangaroos (Fig: 9.31-a), Koalas (Fig: 9.31-b) and Opossums (Fig: 9.31-c) are examples of pouch mammals. A marsupial (pouch mammal) is born very early in its development and completes its embryonic development while nursing. In most species, the nursing young are held within a maternal pouch called a marsupium.



Fig: 9.31 (a) Kangaroo Fig: 9.31 (b) Koala bear

Fig: 9.31 (c) Opossums

Sub class Eutheria (Placental mammals)

Eutherian mammals have a longer period of pregnancy. Young eutherians complete their embryonic development within the uterus, joined to the mother by the placenta. Embryo receives oxygen and food from the mother's circulation and discharges the wastes into her blood through the placenta. Common eutherian mammals are Hedge hogs (Fig: 9.32-a), pygmy shrews (Fig: 9.32-b), rats (Fig: 9.32-c), rabbits (Fig: 9.32-d), sheep (Fig: 9.32-e), goats (Fig: 9.32-f), donkey (Fig: 9.32-g), zebra (Fig: 9.32-h) and giraffe (Fig: 9.32-i).

Among the large placental mammals elephants are terrestrial whereas whale the largest of all the animals are aquatic. Bats are the only flying mammals.



Fig: 9.32 (a) Hedgehog



Fig: 9.32 (b) Pygmy shrew



Fig: 9.32 (c) Rat



Fig: 9.32 (d) Rabbit

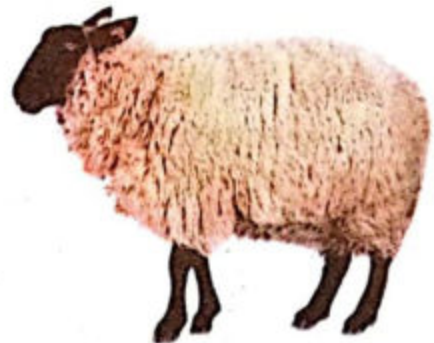


Fig: 9.32 (e) Sheep

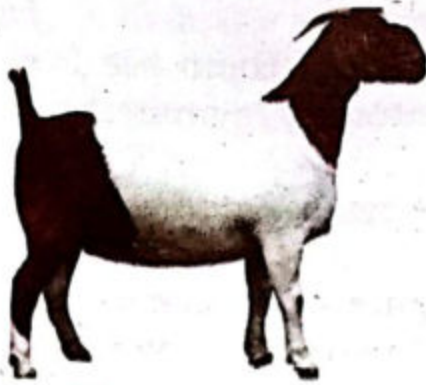


Fig: 9.32 (f) Goat



Fig: 9.32 (g) Donkey



Fig: 9.32 (h) zebra

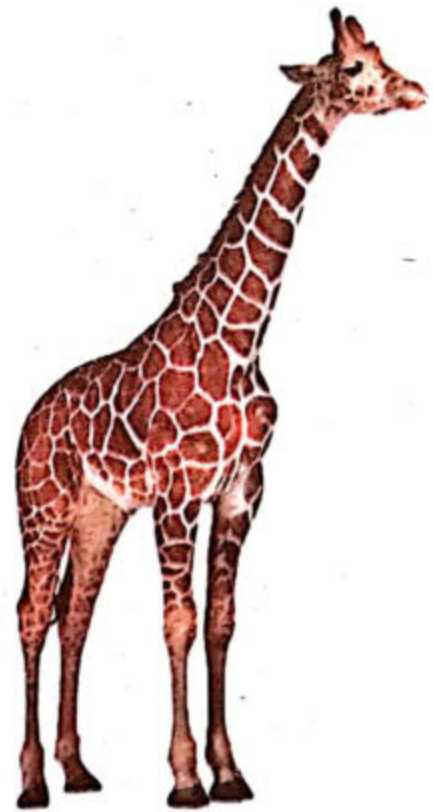


Fig: 9.32 (i) Giraffe

SUMMARY

- Animals are eukaryotic, multicellular heterotrophs (ingestive) which reproduce sexually and are usually motile.
- Animal kingdom is most diversified group as far as number of species is concerned. They range in size from microscopic to very huge organisms.
- Animals are thought to have evolved from volvox like protoctist.
- Kingdom animalia is divided into 33 groups called phyla, on the basis of comparative morphology and internal architecture.
- Most animals have a recognizable symmetry, either radial or bilateral. Sessile animals usually exhibit radial symmetry whereas motile animals are usually bilaterally symmetrical.
- Most animals have a body cavity (coelom) between their body wall and digestive tube.
- Protostome and deuterostomes differ in early embryological development and fate of blastopore.
- Sponges are asymmetrical sessile aquatic filter feeder lack tissue organization.
- Cnidarians are radially symmetrical diploblastic animals with two body forms polyp and medusa.
- Platyhelminthes are commonly called flat worm. They are bilaterally symmetrical triploblastic acoelomates.
- Round worms are the most abundant animals. Most of them are free living whereas many are parasites. They are bilaterally symmetrical triploblastic pseudocoelomates.
- Annelids are segmented worms. They are bilaterally symmetrical coelomate and protostomes.
- Molluscs are bilaterally symmetrical, non-segmented coelomate animals whose body can be divided roughly into a head-foot, a visceral mass and a mantle.
- Arthropoda constitute the most diverse phylum. They are bilaterally symmetrical coelomate and protostomes. Exoskeleton is made up of chitin.
- Echinoderms are radially symmetrical coelomates and deuterostomes. Body is covered over by spiny calcareous plates and thousands of tiny multipurpose tube feet.
- Chordates are the most advanced animals include a small number of invertebrate chordates and vertebrates.
- All Chordates are bilaterally symmetrical animals that are characterized by a notochord, a hollow dorsal nerve cord and pharyngeal gill slits. Many have post anal tail.



EXERCISE

1. Encircle the correct choice.

- (i) Choose the incorrect pair?
 - (a) Sponges → Spicules.
 - (b) Cnidaria → Nematocytes.
 - (c) Segmented worm → Psuedopodia.
 - (d) Arthropoda → jointed legs.
- (ii) Water movement through sponge?
 - (a) Ostia → Spongocoel → Osculum.
 - (b) Spongocoel → Osculum → Ostia.
 - (c) Ostia → Osculum → Spongocoel.
 - (d) Osculum → Spongocoel → Ostia.
- (iii) Choose the term which encompasses all other in the list

(a) Coelomate	(b) Protostome
(c) Bilateria	(d) Triploblastic
- (iv) Dry scaly skin, ectothermic, two pairs of legs, lungs, Internal Fertilization, eggs with amniotic shell, and a three chambered heart are all properties of which class?

(a) Amphibia	(b) Reptilia
(c) Aves	(d) Mammals
- (v) Pick the odd one out?

(a) Star fish.	(b) Brittle star.
(c) Lamprey.	(d) Sea urchin.
- (vi) Which of the following terms or structures is properly associated only with animals?

(a) Genes.	(b) Cell wall.
(c) Autotrophy.	(d) Sexual reproduction.
- (vii) An adult animal that possesses bilateral symmetry is most certainly also

(a) Triploblastic.	(b) Deuterostomes.
(c) Coelomate.	(d) Protostomes.
- (viii) Protostome characteristics generally include which of the following?
 - (a) A Mouth that develops secondly and far away from the blastopore
 - (b) Radial body
 - (c) Radial cleavage.
 - (d) Determinate cleavage
- (ix) Sponges structural materials (spicules) are manufactured by the

(a) Pinacocytes.	(b) Choanocytes.
(c) Amoebocytes.	(d) Porocytes.



(x) Which of the following can be used to distinguish a nematode from annelids?

- I. Type of body cavity
- II. Number of muscle layers in the body wall
- III. Presence of segmentation
- IV. Number of embryonic tissue layers

- (a) I only
- (b) II only
- (c) I and II
- (d) I, II and III

2. Write short answers of the following questions:

1. How animals are classified?
2. Why sponges placed in animal kingdom, when it lacks tissue organization?
3. Why flat worms are named so?
4. Differentiate between invertebrate chordates and vertebrates
5. Differentiate between cartilaginous fishes and bony fishes
6. Why protostomes are named so?
7. Why diploblastic animals are different from the triploblastic?
8. Why echinoderms are placed in bilateria when adults are radially symmetrical?

3. Write detailed answers of the following questions:

1. Discuss the basic factors which helps in classification of animals
2. Write down evolutionary adaptations of phylum annelida.
3. Write down the general characters of phylum arthropoda
4. Describe the general characters of phylum mollusca
5. What are mammals? Name its sub-classes. Give the characters and examples of each sub class.